

Code-Lay■■■out, Readability and Reusability

Date	29 September 2026
Team ID	UNM1659GenieBirds
Project Name	EduGenie – AI-Powered Learning Assistant
Maximum Marks	5 Marks

Code-Layout, Readability and Reusability:

This document evaluates the quality of the EduGenie codebase in terms of structure, readability, maintainability, and reuse. The project is organized into clear functional components for question answering, topic explanation, quiz generation, summarization, and personalized learning-path recommendations.

Code Layout Checklist:

S.No	Code Quality Parameter	Description	Followed (Yes / No / Partial)	Remarks
1	Consistent Indentation	Uniform spacing and indentation are used throughout the code.	Yes	Consistent formatting improves readability and maintenance.
2	Proper File Structure	Files and folders are logically organized according to their responsibilities.	Yes	UI, logic, and supporting resources are kept organized.
3	Meaningful Variable Names	Variables and identifiers clearly reflect their purpose.	Yes	Names are chosen to make the code easy to understand.
4	Function / Method Names	Functions and methods use descriptive names related to their tasks.	Yes	Feature-specific functions improve code clarity.
5	Code Comments	Comments explain important logic where clarification is useful.	Yes	Key processing steps and feature logic are documented.
6	Modular Design	Code is divided into reusable functions/modules for individual features.	Yes	Quiz, summary, explanation, and learning-path logic can be maintained separately.
7	No Redundant Code	Duplicate or unnecessary code is minimized or removed.	Yes	Common operations are reused instead of repeatedly written.
8	Error Handling	Invalid inputs and processing errors are handled appropriately.	Yes	User-friendly handling is provided for incomplete or invalid requests.

Reusable Components / Modules:

S.No	Component / Module Name	Language / Technology	Where Reused	Reusability Level (High / Medium / Low)
1	Question & Answer Module	Python / AI API	Ask a Question feature	High

2	Topic Explanation Module	Python / AI API	Explain a Topic feature	High
3	Quiz Generation & Evaluation	Python / AI API + UI	Generate Quiz feature	High
4	Summary Generator	Python / AI API	Summary feature	High
5	Learning Path Generator	Python / AI API	Recommended Learning Path	High

Overall Code Quality Assessment:

Aspect	Rating (1-5)	Comments
Code Layout & Structure	5	Clear separation of feature components and organized code structure.
Readability	5	Descriptive naming, consistent indentation, and straightforward logic.
Reusability	5	Common feature logic is modular and can be reused or extended.
Documentation / Comments	4	Important logic is documented; additional comments can be added as the project grows.
Overall Score	5	EduGenie follows a clean, maintainable and reusable coding approach.

Team Members: Priyadharshini G (Team Leader) Bushra Fathima T (Team Member) Pooja M (Team Member)